

## SKULTE, Latvia

WMO: 263260

Lat: 57.3006N

Lon: 24.4122E

Elev: 28

StdP: 14.68

Time Zone: 2.00 (EUE)

Period: 95-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -1.1       | 4.6 | -6.8                          | 3.8 | 1.4  | -1.6 | 5.0 | 5.9  | 17.4                  | 33.4 | 16.1 | 33.4 | 4.9                   | 80   | 0.538 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 13.0                   | 82.0            | 67.8 | 78.0 | 66.0 | 74.6 | 64.7 | 70.2                | 78.4 | 67.7 | 74.9 | 65.8 | 72.5 | 6.2                  | 130  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 67.1                            | 99.8 | 74.5 | 64.9 | 92.4 | 71.8 | 63.1 | 86.6 | 69.6 | 34.2          | 78.6 | 32.1 | 74.8 | 30.6 | 72.6 | 76.1              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |  |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|--|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |  |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |  |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |  |
| 15.1              | 13.6 | 12.1 | DB                         | -7.3 | 87.0               | 6.7 | 1.9                                                | -12.2 | 88.4       | -16.1 | 89.6       | -19.9 | 90.7       | -24.7 | 92.1 |  |
|                   |      |      | WB                         | -7.5 | 73.7               | 6.3 | 1.9                                                | -12.1 | 75.1       | -15.8 | 76.2       | -19.3 | 77.3       | -23.9 | 78.7 |  |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 44.6   | 24.9  | 27.2  | 32.4  | 42.2  | 52.1  | 60.1  | 64.7  | 63.1  | 55.5  | 44.8  | 36.9  | 29.8 |
|                                                                                  | DBStd       | 15.77  | 9.89  | 10.13 | 7.20  | 7.53  | 8.54  | 6.37  | 5.69  | 4.82  | 5.70  | 6.42  | 8.19  | 9.27 |
|                                                                                  | HDD50       | 3519   | 778   | 639   | 545   | 256   | 75    | 2     | 0     | 0     | 20    | 184   | 393   | 627  |
|                                                                                  | HDD65       | 7598   | 1243  | 1059  | 1010  | 684   | 409   | 178   | 72    | 95    | 288   | 625   | 843   | 1092 |
|                                                                                  | CDD50       | 1535   | 0     | 0     | 0     | 23    | 140   | 304   | 457   | 405   | 183   | 23    | 0     | 0    |
|                                                                                  | CDD65       | 137    | 0     | 0     | 0     | 0     | 9     | 29    | 64    | 34    | 1     | 0     | 0     | 0    |
|                                                                                  | CDH74       | 866    | 0     | 0     | 0     | 0     | 75    | 198   | 399   | 189   | 5     | 0     | 0     | 0    |
|                                                                                  | CDH80       | 174    | 0     | 0     | 0     | 0     | 8     | 39    | 101   | 26    | 0     | 0     | 0     | 0    |
| Wind                                                                             | WSAvg       | 6.1    | 6.8   | 6.7   | 6.3   | 5.8   | 5.4   | 5.6   | 5.3   | 5.4   | 6.0   | 6.5   | 6.7   | 7.0  |
| Precipitation                                                                    | PrecAvg     | 29.3   | 2.3   | 1.9   | 1.6   | 1.5   | 1.9   | 2.8   | 3.1   | 3.4   | 2.5   | 3.4   | 2.8   | 2.4  |
|                                                                                  | PrecMax     | 38.2   | 4.3   | 3.0   | 3.0   | 3.1   | 3.6   | 7.6   | 6.2   | 6.7   | 5.0   | 7.4   | 4.3   | 3.9  |
|                                                                                  | PrecMin     | 21.3   | 0.8   | 0.4   | 0.5   | 0.4   | 0.6   | 0.8   | 0.3   | 0.0   | 0.5   | 1.7   | 0.4   | 1.3  |
|                                                                                  | PrecStd     | 4.0    | 1.0   | 0.7   | 0.7   | 0.7   | 0.9   | 1.5   | 1.7   | 1.9   | 1.1   | 1.3   | 1.1   | 0.7  |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 41.6  | 44.2  | 54.5  | 69.8  | 81.0  | 84.2  | 86.7  | 83.5  | 73.9  | 62.2  | 51.6  | 44.8 |
|                                                                                  |             | MCWB   | 40.1  | 40.0  | 44.1  | 53.8  | 63.3  | 66.9  | 70.8  | 69.5  | 62.9  | 55.5  | 49.2  | 42.3 |
|                                                                                  | 2%          | DB     | 39.6  | 40.6  | 47.4  | 63.8  | 75.3  | 79.9  | 83.1  | 78.9  | 69.4  | 58.4  | 49.4  | 43.0 |
|                                                                                  |             | MCWB   | 38.2  | 37.8  | 41.0  | 51.8  | 61.2  | 65.3  | 69.5  | 66.3  | 60.5  | 54.5  | 47.0  | 41.1 |
|                                                                                  | 5%          | DB     | 37.9  | 38.6  | 43.4  | 59.2  | 70.2  | 74.7  | 78.2  | 75.5  | 66.1  | 55.7  | 47.7  | 41.8 |
|                                                                                  |             | MCWB   | 36.6  | 36.7  | 39.2  | 48.7  | 57.9  | 64.4  | 68.0  | 64.6  | 59.4  | 52.5  | 45.7  | 40.1 |
|                                                                                  | 10%         | DB     | 36.4  | 36.9  | 40.7  | 55.1  | 66.2  | 70.7  | 74.4  | 72.4  | 63.7  | 53.8  | 46.2  | 40.3 |
|                                                                                  |             | MCWB   | 35.2  | 35.5  | 37.6  | 46.5  | 57.3  | 61.9  | 66.4  | 63.5  | 58.4  | 51.2  | 44.6  | 38.7 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 40.5  | 41.1  | 45.8  | 55.2  | 65.3  | 70.6  | 73.1  | 72.4  | 64.5  | 57.4  | 49.5  | 43.2 |
|                                                                                  |             | MCDB   | 41.3  | 43.6  | 52.0  | 65.4  | 77.8  | 80.1  | 81.9  | 80.0  | 71.4  | 60.5  | 50.8  | 44.0 |
|                                                                                  | 2%          | WB     | 38.5  | 38.4  | 41.8  | 52.3  | 62.8  | 67.0  | 71.2  | 68.2  | 62.0  | 54.8  | 47.3  | 41.7 |
|                                                                                  |             | MCDB   | 39.2  | 40.1  | 46.2  | 62.3  | 71.6  | 75.6  | 79.9  | 75.5  | 66.4  | 57.4  | 48.6  | 42.7 |
|                                                                                  | 5%          | WB     | 36.7  | 36.9  | 39.8  | 49.6  | 60.3  | 64.9  | 69.1  | 66.0  | 60.4  | 53.2  | 46.0  | 40.3 |
|                                                                                  |             | MCDB   | 37.6  | 38.3  | 42.8  | 57.5  | 68.0  | 72.7  | 76.8  | 72.7  | 64.9  | 55.4  | 47.4  | 41.6 |
|                                                                                  | 10%         | WB     | 35.4  | 35.6  | 38.1  | 47.0  | 57.8  | 62.9  | 67.0  | 64.4  | 58.9  | 51.4  | 44.6  | 38.9 |
|                                                                                  |             | MCDB   | 36.4  | 36.7  | 40.4  | 54.1  | 64.5  | 70.3  | 73.4  | 70.9  | 63.2  | 53.5  | 46.2  | 40.3 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 7.8   | 7.6   | 9.8   | 14.0  | 15.4  | 13.0  | 13.0  | 14.2  | 11.6  | 9.1   | 6.1   | 6.0  |
|                                                                                  |             | MCDBR  | 5.3   | 6.4   | 12.8  | 21.2  | 21.5  | 18.7  | 19.3  | 19.3  | 15.4  | 9.6   | 5.6   | 5.2  |
|                                                                                  |             | MCWBR  | 5.1   | 4.8   | 8.2   | 11.5  | 10.6  | 9.2   | 8.8   | 9.6   | 9.4   | 7.1   | 5.6   | 5.1  |
|                                                                                  | 5% WB       | MCDBR  | 5.5   | 6.3   | 11.3  | 18.9  | 18.5  | 16.1  | 16.9  | 15.5  | 13.3  | 8.6   | 5.7   | 5.3  |
|                                                                                  |             | MCWBR  | 5.5   | 5.2   | 7.8   | 10.7  | 9.7   | 9.1   | 8.6   | 9.6   | 9.2   | 7.1   | 5.9   | 5.4  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.290  | 0.318 | 0.347 | 0.384 | 0.376 | 0.375 | 0.406 | 0.402 | 0.368 | 0.345 | 0.331 | 0.289 |      |
|                                                                                  | taud        | 2.178  | 2.179 | 2.200 | 2.272 | 2.375 | 2.415 | 2.348 | 2.360 | 2.432 | 2.406 | 2.253 | 2.175 |      |
|                                                                                  | Ebn at Noon | 178    | 226   | 252   | 262   | 273   | 274   | 262   | 254   | 247   | 218   | 161   | 143   |      |
|                                                                                  | Edh at Noon | 17     | 26    | 33    | 36    | 35    | 34    | 36    | 33    | 27    | 21    | 16    | 13    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 134    | 356   | 795   | 1269  | 1708  | 1784  | 1703  | 1348  | 876   | 413   | 133   | 84    |      |
|                                                                                  | RadStd      | 31     | 70    | 115   | 141   | 164   | 132   | 146   | 130   | 93    | 61    | 20    | 12    |      |

## Historical Trends

|                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                         |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| Regional (37 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page